

Seminar 3.

Def $\forall \varepsilon > 0 \quad \exists N : |a_n - L| < \varepsilon, \quad \forall n > N$

• Sandwich theorem: If $a_n \leq b_n \leq c_n$, and $\lim a_n = \lim c_n = L \Rightarrow \lim b_n = L$

Algebra limit: $\lim a_n = A \quad \lim b_n = B$

$$\lim a_n \pm \lim b_n = A \pm B$$

$$\lim a_n \cdot \lim b_n = A \cdot B$$

$$\frac{\lim a_n}{\lim b_n} = \frac{A}{B}, \quad B \neq 0$$

Standard limits:

- $\lim a^n = 0$, for $0 < |a| < 1$
- $\lim \sqrt[n]{a} = 1$, $a > 0$
- $\lim \sqrt[n]{n} = 1$
- $\lim \frac{n^k}{a^n} = 0$, $a > 1, k \in \mathbb{N}$.

$$\lim n^k \cdot a^n = 0, \quad 0 < a < 1, k \in \mathbb{N}$$

Types of indet. (turn regroup) $\rightarrow \{\infty, \infty\}$

Types of indet. $\rightarrow \{\frac{\infty}{\infty}\}$
 $\Downarrow \frac{1}{\infty} = 0$

$\downarrow \{\frac{0}{0}\}$

$\downarrow \{\infty - \infty\}$

$\downarrow \{1 \cdot \infty\}$

$\downarrow \{0 \cdot \infty\}$

Problem 1.

a) $\lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + n + 5}}{n + 1} = \frac{\sqrt{\infty^2 + \infty + 5}}{\infty + 1} = \left\{ \frac{\infty}{\infty} \right\} =$

$= \left[\sqrt{n^2 + n + 5} = \sqrt{n^2 \left(1 + \frac{1}{n} + \frac{5}{n^2} \right)} = n \sqrt{1 + \frac{1}{n} + \frac{5}{n^2}} \right] =$

$= \lim_{n \rightarrow \infty} \frac{n \sqrt{1 + \frac{1}{n} + \frac{5}{n^2}}}{n + 1} = \left[n + 1 = n \left(1 + \frac{1}{n} \right) \right] =$

$\frac{\cancel{n} \sqrt{1 + \frac{1}{n} + \frac{5}{n^2}}}{\cancel{n} \left(1 + \frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{n} + \frac{5}{n^2}}}{1 + \frac{1}{n}} =$

$$= \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{n} + \frac{5}{n^2}}}{1 + \frac{1}{n}} \quad \left[\frac{a}{\infty} = 0 \right] \quad \lim_{n \rightarrow \infty} 1 + \frac{1}{n} = 1.$$

$$b) \lim_{n \rightarrow \infty} \frac{(-3)^n + 7^{n+1}}{(-3)^{n+2} + 7^n} = [a^{b+c} = a^b \cdot a^c] =$$

$$= \lim_{n \rightarrow \infty} \frac{(-3)^n + 7 \cdot 7^n}{9(-3)^n + 7^n} = \left[\begin{array}{l} \text{dividing on fastest} \\ \text{growth function: } 7^n \end{array} \right]$$

$$\left[\begin{array}{l} \lim_{n \rightarrow \infty} (a)^n = 0 \\ 0 < |a| < 1 \\ -1 < a < 1 \end{array} \right]$$

$$\lim_{n \rightarrow \infty} \frac{(-\frac{3}{7})^n + 7}{9 \cdot (-\frac{3}{7})^n + 1} = 7.$$

$$c) \lim_{n \rightarrow \infty} \frac{\sqrt{5n^2 + \sqrt{n^3 + 5n}} - n + 3}{\sqrt{n^4 - 3n + 7} + \sqrt{n-1}} = \left\{ \frac{\infty}{\infty} \right\}$$

$$\begin{aligned} \sqrt{n^3 + 5n} &= \sqrt{n^2(n + \frac{5}{n})} = n \sqrt{n + \frac{5}{n}} \\ \sqrt{n^4(\frac{1}{n} + \frac{5}{n^3})} &= n^2 \sqrt{\frac{1}{n} + \frac{5}{n^3}} \end{aligned} \quad (n + \frac{5}{n})^{\frac{1}{2}}$$

$$\bullet \sqrt{n^4 - 3n + 7} = \sqrt{n^4(1 - \frac{3}{n^3} + \frac{7}{n^4})} = n^2 \sqrt{1 - \frac{3}{n^3} + \frac{7}{n^4}}$$

$$\bullet \sqrt{n-1} = \sqrt{n^2(\frac{1}{n} - \frac{1}{n^2})} = n \sqrt{\frac{1}{n} - \frac{1}{n^2}}$$

$$\lim_{n \rightarrow \infty} \frac{5n^2 + n^2 \sqrt{\frac{1}{n} + \frac{5}{n^3}} - n + 3}{n^2 \sqrt{1 - \frac{3}{n^3} + \frac{7}{n^4}} + n \sqrt{\frac{1}{n} - \frac{1}{n^2}}} =$$

$$= \lim_{n \rightarrow \infty} \frac{5n^2 - n + 3}{n^2} = \lim_{n \rightarrow \infty} \frac{5 - \frac{1}{n} + \frac{3}{n^2}}{1} = 5.$$

$$d) \lim_{n \rightarrow \infty} \frac{n^2 - 2n + 5}{n^2 - 2n + 5} = \{ \infty - \infty \} = \left[\begin{array}{l} n^2 - 2n + 1 + 4 = \\ = (n-1)^2 + 4 \end{array} \right]$$

$$\lim_{n \rightarrow \infty} \underbrace{(n-1)^4}_{\infty} + 4 = +\infty \quad (\text{DNE})$$

$$e) \lim_{n \rightarrow \infty} \frac{5n^2 + n + 2}{1 + 2 + \dots + n} = \left[\begin{array}{l} 1+2+\dots+n = \text{arithmetic prog.} = \\ = \frac{n(n+1)}{2} \end{array} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{10n^2 + 2n + 4}{n^2 + n} = \lim_{n \rightarrow \infty} \frac{10 + \frac{2}{n} + \frac{4}{n^2}}{1 + \frac{1}{n}} = 10.$$

$$f) \lim_{n \rightarrow \infty} \frac{n^3 + 3^n}{n + 3^{n+1}} = \lim_{n \rightarrow \infty} \frac{n^3 + 3^n}{n + 3 \cdot 3^n} \cdot \left[\begin{array}{l} 3^n \text{ growth} \\ \text{faster than } n^3 \end{array} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{n^3}{3^n} + 1}{\frac{n}{3^n} + 3} \cdot \left[\begin{array}{l} \frac{n}{3^n} \xrightarrow{\text{known value}} 0 \\ \frac{n^3}{3^n} \xrightarrow{\text{known value}} 0 \end{array} \right] = \frac{1}{3}$$

$$g) \lim_{n \rightarrow \infty} \sqrt[n]{3^n + n \cdot 2^n} = \lim_{n \rightarrow \infty} \sqrt[n]{3^n \left(1 + n \cdot \left(\frac{2}{3} \right)^n \right)} =$$

$$= \lim_{n \rightarrow \infty} \sqrt[n]{3^n} = 3$$

Problem 5.

$$\lim_{n \rightarrow \infty} \left(\sqrt{n+2025} - \sqrt{n} \right) = \{ \infty - \infty \} =$$

$$= \lim_{n \rightarrow \infty} \frac{(\sqrt{n+2025} - \sqrt{n})(\sqrt{n+2025} + \sqrt{n})}{\sqrt{n+2025} + \sqrt{n}} =$$

$$= \lim_{n \rightarrow \infty} \frac{2025}{\sqrt{n+2025} + \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{\frac{2025}{\sqrt{n}}}{1 + 1} = 0$$

$$\begin{array}{l} \frac{2025}{\sqrt{n+2025} + \sqrt{n}} \\ \downarrow \frac{1}{2} \quad \downarrow \frac{1}{2} \\ (n+2025)^{\frac{1}{2}} \quad n^{\frac{1}{2}} \\ 1 + \dots 0 \quad \frac{2025}{n^{\frac{1}{2}}} \quad 1 \end{array}$$

$$\sqrt{1 + \frac{2025}{n}} - \sqrt{1} \cdot \sqrt{1 + \frac{2025}{n}}$$

$$\sqrt{n(n+1)} - n = \sqrt{n^2 + n} - n$$

Problem 6.

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} \right)$$

$$\sum_{k=1}^n \frac{1}{k(k+1)}$$

$$\frac{1}{k(k+1)} = \frac{A^{(k+1)}}{k} + \frac{B^{(k)}}{k+1} = \frac{Ak + A + Bk}{k(k+1)}$$

$$Ak + A + Bk = 1$$

$$\begin{cases} (A+B)k = 0 \\ A = 1 \end{cases}$$

$$\frac{1}{n} - \frac{1}{n+1} \xrightarrow{-\frac{1}{2}}$$

$$\Rightarrow \begin{matrix} A=1 \\ B=-1 \end{matrix} \quad \frac{1}{1} - \frac{1}{1+1} + \frac{1}{2} - \frac{1}{2+1} + \dots$$

$$\textcircled{1} - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \frac{1}{4} + \dots + \frac{1}{n} - \textcircled{\frac{1}{n+1}} =$$

$$= \frac{1 - \frac{1}{n+1}}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1.$$